AMGPRA[1]: implementation, reproducibility, and application to a real-world problem

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[1] C. Zhang, C. Song, and A. Shafieezadeh, *Adaptive Reliability Analysis for Multi-fidelity Models using a Collective Learning Strategy*, in "Structural Safety", vol.94, 2022.

Overview

1. Introduction
   - Methodological recap
   - Active learning methods
2. The AMGPRA method explained
3. Implementation
4. Reproducibility of the results
5. Case study: `solar`
   - Problem physics and modeling
   - From optimization to reliability
6. Conclusion

Introduction and reminders

The reliability analysis framework

Goal: quantify the probability that a system fails, *i.e.* that the **limit-state function** $g : \mathcal{X} \to \mathbb{R}$ becomes negative:

$$P_f = \mathbb{P}\big(g(\mathbf{x}) \leq 0\big),\ \mathbf{x} \in \mathcal{X} = \mathbb{R}^d$$

Context: in engineering, $g(\mathbf{x})$ is often

- computationally expensive
- dependent on several uncertain inputs

Challenge: Estimating P_f accurately while limiting the number of expensive simulations

Challenges and perspectives

1. Direct Monte Carlo Simulation (MCS) ?

- Simple and exact
- But → Requires $10^5 \sim 10^6$ calls to g
- **Infeasible** if $g(\mathbf{x})$ is costly (minutes/hours per call)

2. Active learning idea

- Replace g by a (cheaper) **GP surrogate** $\hat{g}$
- Smart sampling → query g only where uncertainty is high
- Focus on the **limit-state boundary**

3. Evolution of active learning methods

Method	Strategy	Limitation
AK-MCS (2011)	Single HF surrogate	All new points are expensive
mfEGRA (2020)	MF: point → then fidelity	Suboptimal (two-step) selection
AMGPRA (2022)	**Joint point + fidelity**	**Cost-normalized CLF**

Recall: surrogate modeling for active learning

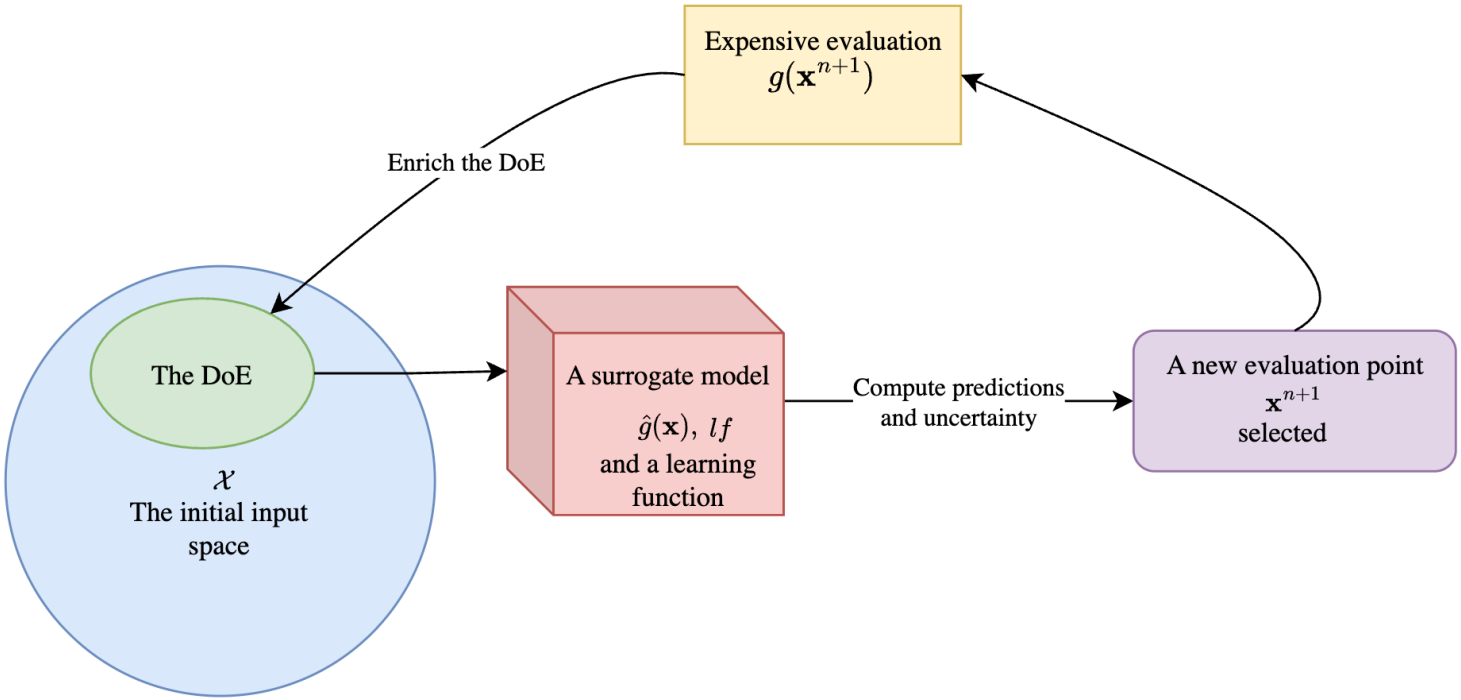


Figure: An active learning procedure using a surrogate model

Example of metamodel: *Gaussian process (GP) regression, or Kriging*

The AMGPRA method explained

AMGPRA in a nutshell

Multi-fidelity GP model: $K+1$ fidelity levels, high-fidelity indexed $l=0$:

$$\hat{g}(0,\mathbf{x}) \sim \mathcal{GP}(\mu_0, \Sigma_0), \qquad \delta_l(\mathbf{x}) = \hat{g}(l,\mathbf{x}) - \hat{g}(0,\mathbf{x}) \sim \mathcal{GP}(0, \Sigma_l).$$

Collective Learning Function (CLF) — the core novelty:

$$\mathrm{CLF}(\mathbf{x}^{n+1}, l_F) = \frac{1}{N_c\, c_l} \sum_{i=1}^{N_c} \Big[lf\big(\mu_P, \sigma_P^2(\mathbf{x}^{(i)})\big) - lf\big(\mu_P, \sigma_F^2(\mathbf{x}^{(i)} \mid \mathbf{x}^{n+1}, l_F)\big) \Big]$$

- **Simultaneously** selects the next point $\mathbf{x}^{n+1}$ *and* fidelity level l_F by maximizing CLF
- Accounts for the **global** variance reduction across the candidate population, normalized by evaluation cost c_l
- Learning function lf can be the EFF or the U-function

The AMGPRA procedure

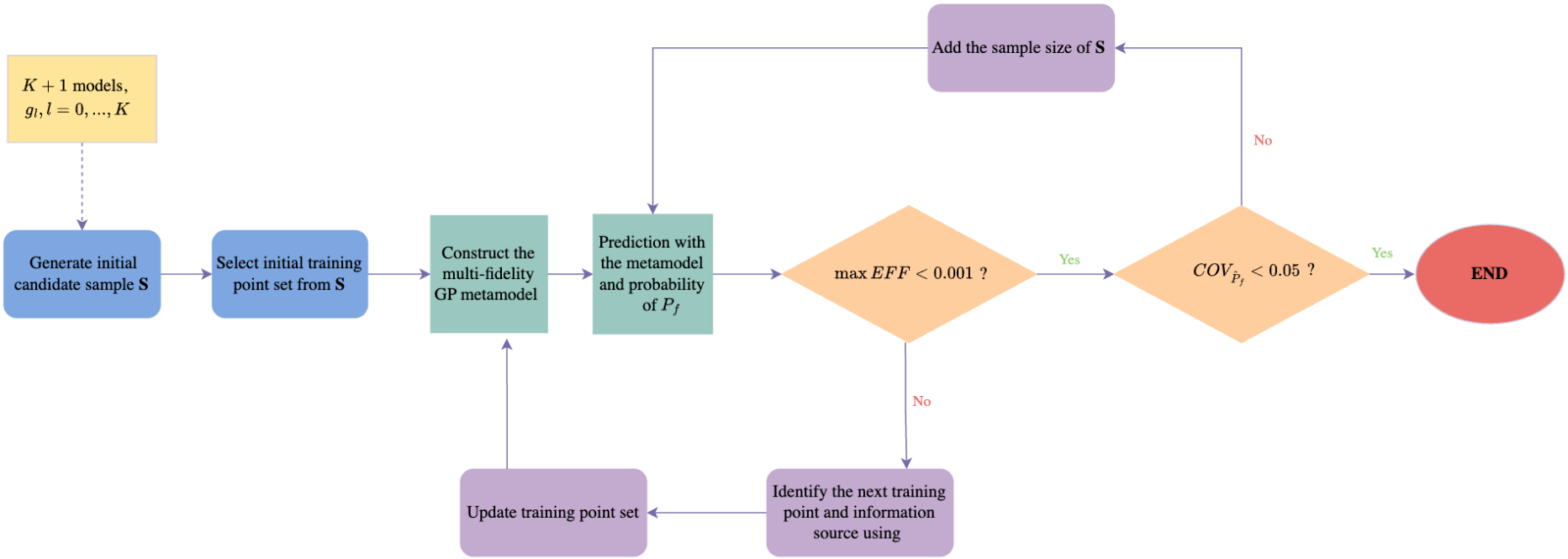


Figure: Flowchart of the AMGPRA approach

Implementation: developing the AMGPRA method from scratch

Reconstructing AMGPRA and mfEGRA from scratch

No public code exists for either method ⇒ both algorithms **fully re-implemented** in Python, using the `SMT` (Surrogate Modeling Toolbox) library as backbone

Surrogate engine → SMT's `MFK` model

Kennedy & O'Hagan (2000) autoregressive formulation.

$$Z_{\mathrm{HF}}(\mathbf{x}) = \rho(\mathbf{x})\, Z_{\mathrm{LF}}(\mathbf{x}) + \delta(\mathbf{x})$$

Software architecture:

Component	Role
`AMGPRA` / `MFEGRA`	Active loop, enrichment, and stopping criteria
`MultiFidelityModel`	Isolation of `SMT` interface (wrapper)
`LearningFunctions`	EFF, U-function, and CLF proxy logic

AMGPRA software architecture - UML

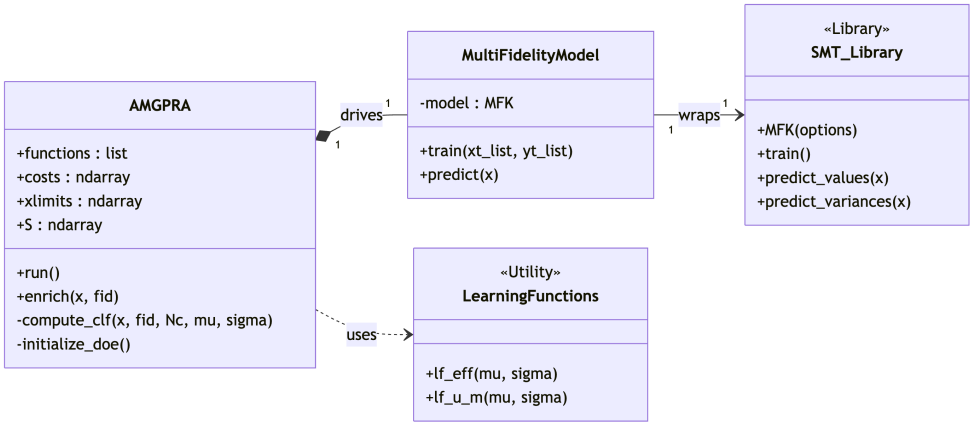


Figure: UML class diagram of the AMGPRA implementation

- `AMGPRA`: active loop (`run()`, `enrich()`), holds the DoE, the cost vector, and S
- `MultiFidelityModel`: *wraps* SMT's `MFK` $\rightarrow$ all calls to `predict_values` and `predict_variances`
- `LearningFunctions`: `lf_eff` and `lf_u_m` called by `_compute_clf` to evaluate the variance reduction proxy

Implementation challenges

1) Covariance unavailability in `MFK`

CLF requires the *posterior covariance* between arbitrary fidelity levels:

$$\Sigma_P\big((l,\mathbf{x}),(l',\mathbf{x}')\big) = \Sigma_{\mathrm{pr}} - \Sigma_{\mathrm{pr}}\,\mathbf{K}^{-1}\,\Sigma_{\mathrm{pr}}$$

- **In AMGPRA**: CLF approximated using **marginal variances** only $\rightarrow$ simplifies computation and avoids internal MFK limits
- **In mfEGRA**: Full reconstruction from kernel hyperparameters via Cholesky factorization (`_post_cov`)

2) HF over-selection in mfEGRA

The criterion favored the High-Fidelity (HF) level too frequently. We introduced an **empirical offset** for stabilization:

$$\mathrm{Var}_P\big(\hat{g}_0(\mathbf{x}^{n+1})\big) \;\leftarrow\; \mathrm{Var}_P\big(\hat{g}_0(\mathbf{x}^{n+1})\big) + 0.3$$

$\Rightarrow$ Penalizes HF selection and encourages cheaper low-fidelity explorations

Reproducibility: towards the 5.1. example from the original paper[2]

[2] Zhang et al. (2022)

Benchmark setup: a tri-fidelity analytical example

Input distribution: $\mathbf{x} = (x_1, x_2) \sim \mathcal{N}_2\big((1.5, 2.5)^\top, \mathbf{I}_2\big)$

HF limit-state function:

$$g_0(x_1, x_2) = \frac{(x_1^2 + 4)(x_2^2 - 1)}{20} - \sin\Big(\frac{5x_1}{2}\Big) - 2$$

LF models: $g_l(\mathbf{x}) = g_0(\mathbf{x}) + \delta_l(\mathbf{x}),\ l = 1, 2$ (additive corrections with aligned zero contours)

Reproduction protocol:

- 5 independent runs per method, same convergence criteria as the paper
- Reference: $\hat{P}_f = 0.0313$ (MCS with 10^6 samples)
- Comparison includes: AK-MCS, mfEGRA, AMGPRA-LF-EFF, AMGPRA-LF-Um.

Reproducibility results

Table: Average results over 5 runs — **ours** vs. paper reference

Method	Source	Avg. Cost	Avg. $\hat{P}_f$	ARE (%)
MCS (Reference)	Paper	10^6	0.0313	–
AK-MCS-EFF	**Ours**	34.0	0.0313	0.040
mfEGRA	Paper	12.87	0.0311	0.020
	Ours	**15.96**	**0.0312**	**0.017**
AMGPRA-LF-EFF	Paper	12.32	0.0310	0.020
	Ours	**15.18**	**0.0313**	**0.028**
AMGPRA-LF-Um	Paper	12.41	0.0312	0.030
	Ours	**14.27**	**0.0312**	**0.021**

- $\hat{P}_f$ **very close to the reference** ($< 0.03\%$ error)
- **Cost $\approx$ +20% vs. Paper**: Likely due to SMT's *autoregressive* MFK vs. original *additive* model + different initial DoE sampling
- **Hierarchy preserved**: AMGPRA ($14 \sim 15$) $<$ mfEGRA (16) $<$ AK-MCS (34)

Visual validation of surrogate quality

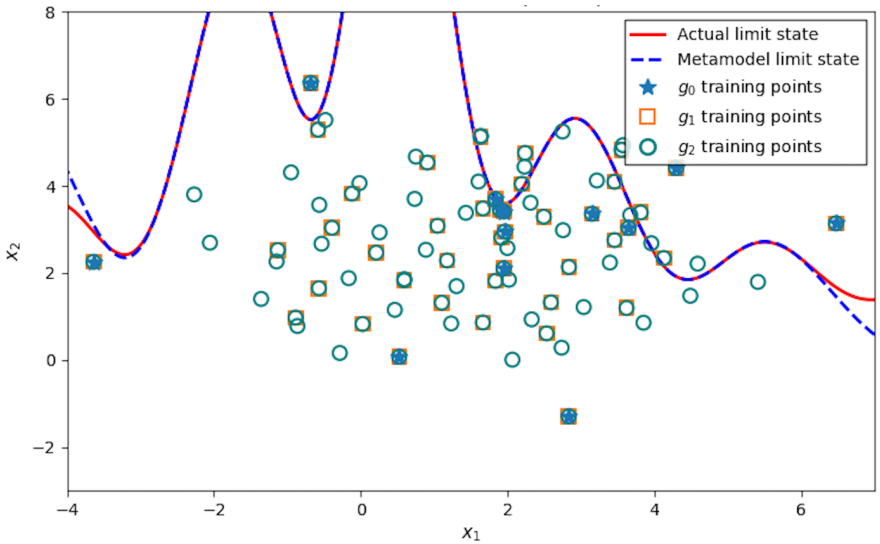


Figure: AMGPRA-LF-EFF - converged at iteration 62 (run 5)

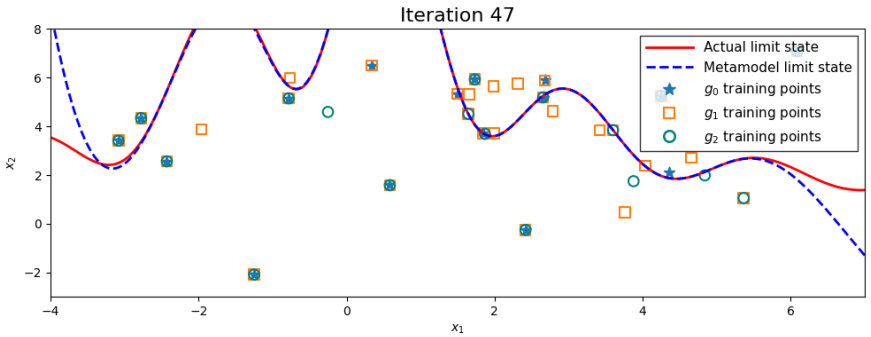


Figure: mfEGRA - converged at iteration 47 (run 5)

- Both methods correctly identify the multi-modal limit-state boundary $\{g_0 = 0\}$
- Validation achieved with a modest number of HF evaluations

Robustness study: what if LF contours are shifted? (1/2)

Motivation: Reference benchmark *favorable by design* → the LF zero contours coincide exactly with the HF boundary ⇒ *how robust are AMGPRA and mfEGRA when this alignment is broken ?*

Modified LF models (same g_0, same input distribution):

$$g_1 = (g_0 + 0.30) \cdot \left[1.6 + 0.3 \sin\left(\frac{5x_1}{22} + \frac{5x_2}{44} + \frac{5}{4}\right) \cos(x_1 + x_2 + 2)\right]$$

$$g_2 = (g_0 - 0.20) \cdot \left[1.4 + 0.25 \sin\left(\frac{5x_1}{22} + \frac{5x_2}{44} + \frac{5}{4}\right) \cos(x_1 + x_2 + 2)\right]$$

- Zero contour of $g_1 \Leftrightarrow g_0 = -0.30$ (shift inward)
- Zero contour of $g_2 \Leftrightarrow g_0 = +0.20$ (shift outward)

LF models remain *correlated* with HF but provide a **biased** indication of the true failure boundary

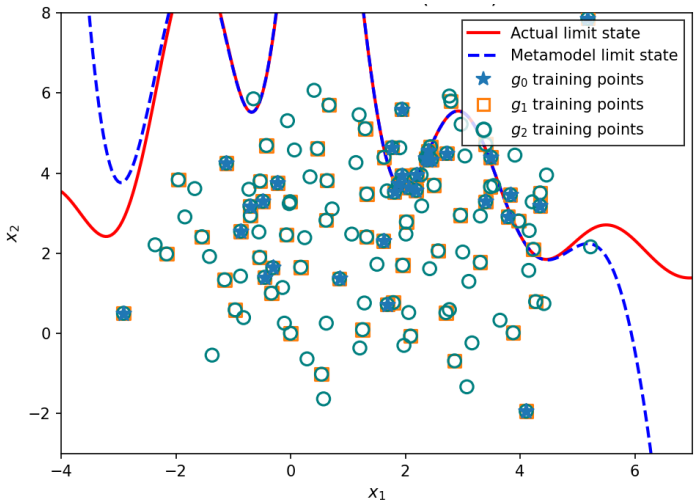
Robustness study: impact of shifted contours (2/2)

Table: Comparative results - original vs. modified (shifted)

Method	Avg. Cost	Avg. $\hat{P}_f$	ARE (%)
Original setting (Reference)			
mfEGRA	15.96	0.0312	0.017
AMGPRA-LF-EFF	15.18	0.0313	0.028
Modified setting (shifted contours)			
mfEGRA	**33.58**	0.0310	0.021
AMGPRA-LF-EFF	71.08	0.0320	0.045
AMGPRA-LF-Um	72.94	0.0314	0.034

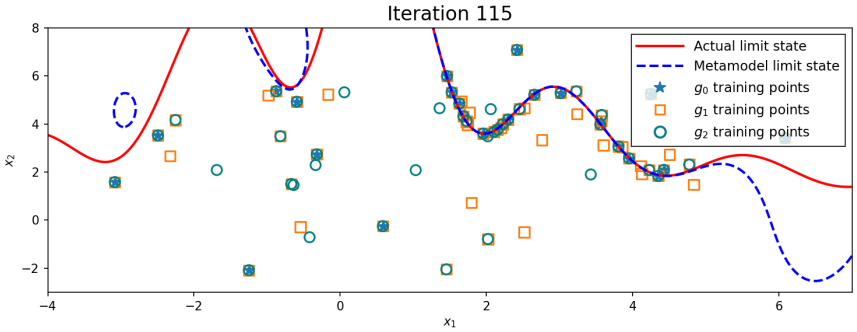
- **AMGPRA cost explodes ($\times 5$)** when contours are shifted, mfEGRA's doubles
- **Accuracy remains high**: All methods are robust in terms of $\hat{P}_f$
- **Ranking reversal**: mfEGRA becomes the cheapest option when the "additive" discrepancy assumption is violated.

Robustness study: impact of shifted contours (2/2)



AMGPRA

(Iteration 122)



mfEGRA

(Iteration 115)

- Reconstruction less accurate than the reference case
- Spurious contours appear in low-probability regions due to the **shifted low-fidelity bias**

Case study: The solar optimization benchmark[a]

[a] Audet et al., *solar: A solar thermal power plant simulator for blackbox optimization benchmarking, 2024.*

Applying AMGPRA to an industrial CSP plant simulator

`solar`: simulator of a concentrated solar power (CSP) plant

- **Heliostats:** Sun-tracking mirrors
- **Receiver:** Captures solar flux to heat molten salt
- **Storage:** Thermal tanks for deferred energy use

Figure: The Thémis CSP tower plant in Targasonne, France

→ High-dimensional blackbox where each evaluation involves expensive ray-tracing and thermodynamic simulations

The solar 10.1 instance

1) System inputs -5 continuous variables

x_1:	H_{hot}	$[1, 50]$ m	x_4:	$T_{\text{rcv}}^{\text{out}}$	$[793, 995]$ K
x_2:	d_{hot}	$[1, 30]$ m	x_5:	$T_{\text{cold}}^{\min}$	$[495, 650]$ K
x_3:	t_{ins}	$[0.01, 2]$ m	Optimum:	$F(\mathbf{x}_{\text{opt}}) \approx 42.42$	

2) Input uncertainty model

- $X_i \sim \mathcal{N}(x_{\text{opt},i},\, \sigma_i^2)$
- **1% variation**: $\sigma_i = 0.01 \times x_{\text{opt},i}$
- Parametric noise around the COBYLA operating point

3) From blackbox optimization to reliability analysis

Cost function with penalty cliffs

$$F(\mathbf{x}) = h_1(\mathbf{x}) + \frac{10^6}{2} \sum_{i=1}^{6} w_i g_i(\mathbf{x})^2$$

(steep gradients at feasibility boundaries)

Reliability problem

$$g(\mathbf{x}) = \tau - F(\mathbf{x})$$
$$P_f = \mathbb{P}(F(\mathbf{X}) > \tau)$$

Fragility thresholds: $\tau(\alpha) = F(\mathbf{x}_{\text{opt}})(1 + \alpha)$,
with $\alpha \in \{0.1\%, 1\%, 5\%, 10\%, 20\%, 30\%, 60\%\}$ the budget overruns

Fidelity calibration: choosing ϕ_{LF}

Preliminary benchmark using mfEGRA at four levels $\phi \in \{0.25, 0.50, 0.75, 1.0\}$:

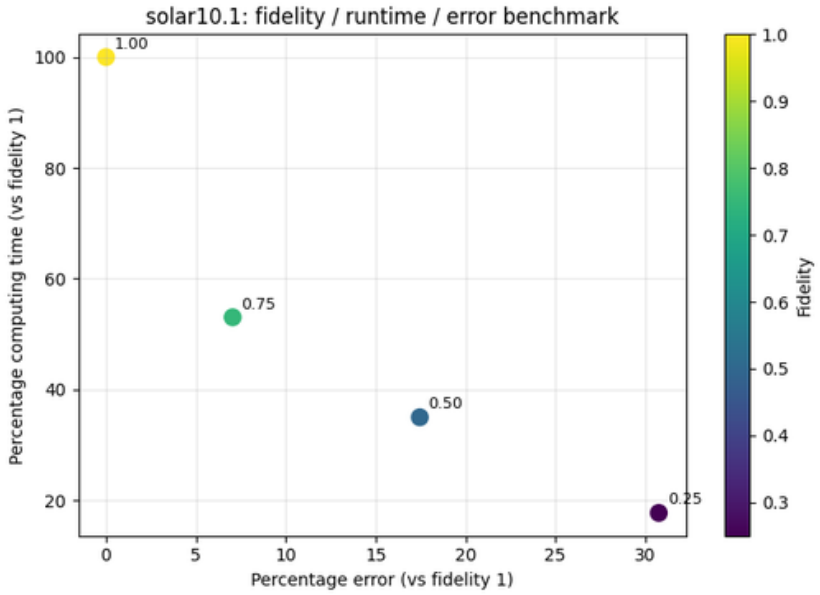


Figure: Trade-off: error vs. computing time fraction

ϕ	Rel. error	Time frac.
0.25	$\approx 31\%$	$\approx 20\%$
0.50	$\approx 18\%$	$\approx 35\%$
0.75	$\approx 7\%$	$\approx 53\%$
1.00	0%	100%

$\phi = 0.5$ occupies the best trade-off: enough accuracy to guide acquisition, at ≈65% cost savings ⇒ **Bi-fidelity setup**: $(\phi_{LF}, \phi_{HF}) = (0.5, 1.0)$

Fragility curve: statistical results and robust zone

AMGPRA estimations for various α

α (%)	Threshold τ	$\hat{P}_f$
0.1	42.46	3.95×10^{-2}
1	42.84	7.20×10^{-2}
5	44.54	1.28×10^{-2}
10	46.66	1.00×10^{-3}
20	50.90	≈ 0
30	55.14	≈ 0
60	67.87	≈ 0

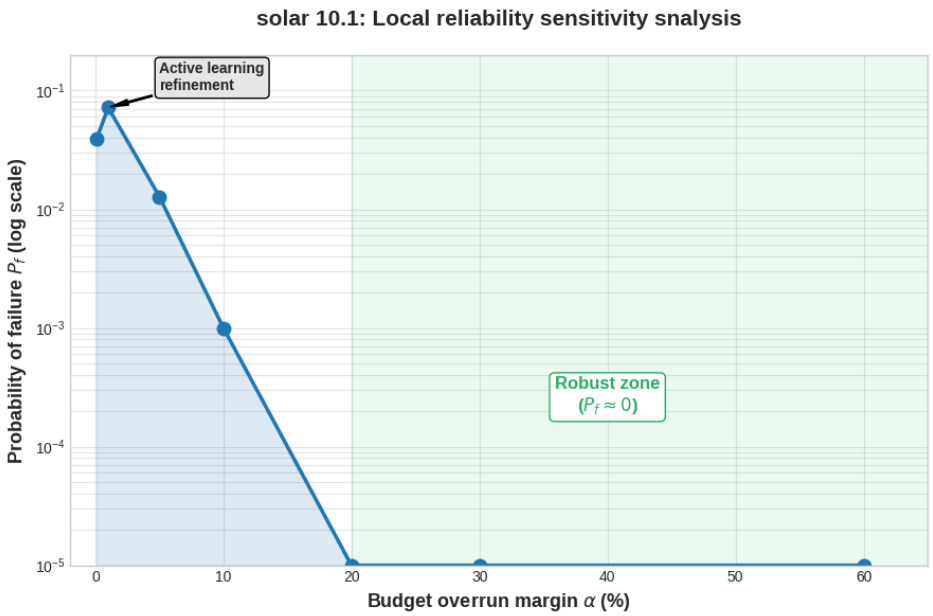


Figure: Fragility curve (log scale)

Green shaded area = **robust zone**

Reference optimum $F(\mathbf{x}_{\text{opt}}) \approx 42.42$

Interpreting the fragility curve

1. Active learning correction

At $\alpha = 0.1\%$ **vs.** 1%: P_f increases due to **surrogate refinement** $\rightarrow$ U-function initially explores the peak of the input distribution, then stabilizes the boundary classification

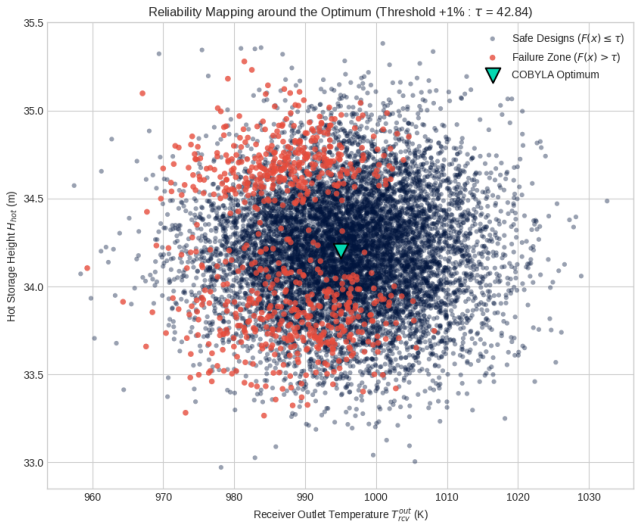
2. Physics of the solar plant

Rapid decay (1% to 10%): Penalty cliffs create steep cost gradients $\rightarrow$ AMGPRA highly efficient here: a few cheap LF evaluations accurately locate this narrow boundary

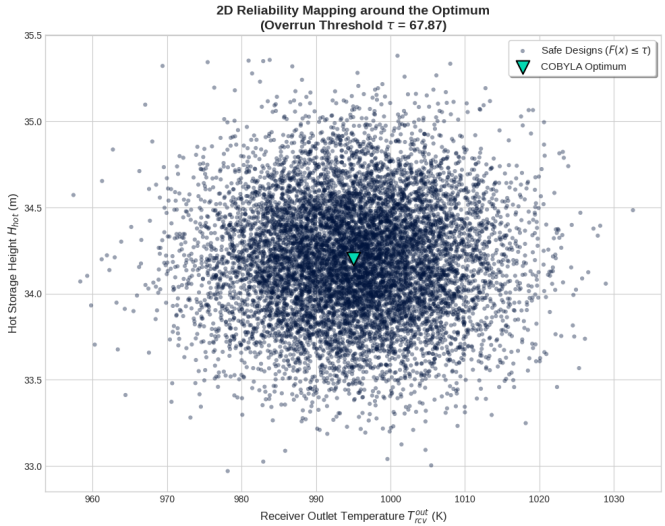
3. The robust zone

For $\alpha \geq 20\%$, risk is negligible under 1% parameter uncertainty $\rightarrow$ A **20% budget margin** guarantees a robust design

2D reliability mapping around the optimum



$\alpha = 1\%$ **threshold** ($\hat{P}_f \approx 7.2\%$)
Failures (red) cluster at low T_{rcv} and low H_{hot}



$\alpha = 60\%$ **threshold** ($\hat{P}_f = 0$)
All 10,000 samples are safe $\rightarrow$ the **robust zone**

Asymmetric sensitivity

- Projection onto ($T_{\mathrm{rcv}}^{\mathrm{out}}$, H_{hot}) reveals that failure is **directional**
- Risk increases when operating below nominal thermal capacity
- This confirms the **asymmetric penalty structure** of the `solar 10.1` cost function $F(\cdot)$

Conclusion and perspectives

Summary and limitations

What we achieved

- **Implementation**: AMGPRA, mfEGRA, and AK-MCS developed using SMT's `MFK`
- **Validation**: Paper cost hierarchy and $\hat{P}_f$ accuracy successfully recovered
- **Sensitivity analysis**: Highlighted AMGPRA's dependency on low-fidelity quality
- `solar`: Application of AMGPRA to a 5D industrial simulator $\rightarrow$ identified a **20% robustness zone**

Main limitations

- **CLF proxy**: Use of marginal variances instead of full posterior covariance (potential sub-optimality in non-additive cases)
- **Heuristics**: `solar` allocation rules and stopping criteria remain empirical
- **Scalability**: Performance in high dimensions ($d \gg 5 - 10$) remains to be tested

Perspectives

1. **Alternative acquisition criteria:** entropy-based or stepwise-uncertainty-reduction criteria may be less sensitive to LF boundary mismatch than EFF/CLF
2. **Formal fidelity allocation:** replace the heuristic variance-threshold rule for `solar` by the cost-optimal policy derived in Zhang et al. (2022)
3. **Confidence bands:** propagate metamodel uncertainty to obtain credible intervals on $\hat{P}_f(\alpha)$, making the fragility curve industrially actionable
4. **High-dimensional extension:** integrate dimensionality reduction (PLS-Kriging, active subspaces) to push AMGPRA beyond $d \approx 10$

Thank you!

Questions?

References

- C. Zhang, C. Song, A. Shafieezadeh, *Adaptive Reliability Analysis for Multi-Fidelity Models using a Collective Learning Strategy*, Structural Safety, vol. 94, 2022.
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- C. Audet et al., *solar: A solar thermal power plant simulator for blackbox optimization benchmarking*, arXiv:2406.00140, 2024.

AMGPRA as an algorithm I

Algorithm 1:

Input: Initial DoE $\mathcal{X}_0$, Monte Carlo sample size N_{MCS}
Output: Estimated probability of failure $\hat{P}_f$

```
procedure AMGPRA(X_0):
    X ← X_0, Generate S, Train multi-fidelity GP ĝ on X
    while True do
        Compute P̂_f, Err_max and COV_P̂f on S using ĝ
        if Err_max ≥ 0.001 then
            Select (x_new, l_new) via the CLF
            if l_new = 0 then
                Add x_new to all fidelities in X
            Retrain ĝ on updated X
        else if COV_P̂f > 0.05 then
            Generate ΔS and update S ← S ∪ ΔS
        else
            return P̂_f
```